

Project Progress - Week 3 of March 2025

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March 21, 2025

This is a file that contains the tracking of the activities related to the OFC project during the third week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the third week of March 2025. The key achievements include:

- The management of Python modules and compilation of LaTeX projects has been learned.
- A first approach to C++ modular development of the functionalities has been initiated.
- The reading of Draine's paper on radiative corrections in polarizability [1] has been initiated.
- The difficulties in implementing C++ modules have been discussed with Raúl. And intuition on how to proceed has been provided.
- The need to use the Discrete Dipole Approximation (DDA) for the calculation of optical forces for particles of size comparable to the wavelength has been discussed.

So far, the project is developing in some areas:

- The development of the algorithm to obtain the polarizability of the particles.
- The modular development, including makefile documentation and Raul's cpp repository and notes analysis.
- The development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.

2 Potential Tasks for the Week

The following tasks are proposed for the week of March 10th to March 15th, 2025:

- Exploration of CMakefile, include directory, and modular development (Notion)

- Develop the algorithm to obtain the polarizability of the particles.
- Implement the calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- Documentation of quasi-static polarizability theory.
- Continue reading Draine's paper on radiative corrections in polarizability [1].
- Analysis of Raúl's `cpp-template-project` repository.
- Check of Raúl's class notes.
- Discuss the progress and difficulties with Raúl.

3 Week Progress

3.1 Monday, March 17th, 2025

The tasks for the week have been defined and the progress of the project has been analyzed. CMakeFiles and include directories have been explored, creating an initial structure (it doesn't compile yet) and notion pages for registering information about the template project `cpp-template-project`.

3.2 Tuesday, March 18th, 2025

The compilation issues with the CMakeFiles have been solved and the environment dependencies had been installed. The first test for optical forces calculation function has been implemented and tested. The function is able to calculate the optical forces given a medium permittivity, a set of dipolar momenta and a set of electric field gradients. A library for dielectric coefficients has been created. In it, a class for permittivities has been defined. A method for gold particles testing has been implemented, but yet not tested correctly.

3.3 Wednesday, March 19th, 2025

The wavelength dependence of gold's permittivity has been checked, yet not graphed explicitly. The compilation issues in VSCode have been solved, apparently related with a compilation file that needs to be created for the clangd extension. The test for gold permittivity has been implemented and tested. The permittivity of gold has been implemented as a function of the wavelength. A first try for cpp executable compiling with the makefile has been done, but it doesn't compile yet. There is an error related with the definition of a permittivity object.

4 Summary

The project has made significant progress during the third week of March 2025. The key achievements include:

- The exploration of CMakeFiles and include directories.

- The implementation of the optical forces calculation function.
- The creation of a library for dielectric coefficients.
- The implementation of a method for gold particles testing.
- The checking of the wavelength dependence of gold's permittivity.
- The solving of compilation issues in VSCode.

5 Difficulties

The main difficulties encountered during the week are:

- The compilation issues in VSCode.
- The definition of a permittivity object.
- The creation of a compilation file for the clangd extension.
- The connection with refractiveindex.info database.
- The creation of a python interface for the cpp library.

6 Current Status of the Project

So far, the project is developing in some areas:

- The development of the algorithm to obtain the polarizability of the particles.
- Implement radiative correction and graphing of gold's permittivity.
- Continue with the development of the algorithm to obtain the dipolar momenta of the particles using gold's basic formula.
- The development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- The creation of a library for dielectric coefficients.
- The creation of a python interface for the cpp library.

7 Next Steps

The next steps for the project are:

- Check the issue with the definition of a permittivity object.
- Check the generation of c++ executables.
- Continue reading Draine's paper on radiative corrections in polarizability [1].

- Write documentation about the polarizability theory (review Raúl’s PhD thesis repository structure).
- Continue with the development of the algorithm to obtain the polarizability of the particles.

References

- [1] B. T. Draine. “The Discrete-Dipole Approximation and Its Application to Interstellar Graphite Grains”. In: *The Astrophysical Journal* 333 (Oct. 1988), p. 848. DOI: 10.1086/166795.